LECTURE 10

6.8. **Fields as vector spaces.** We want to show that any finite field F has p^n elements, where p is a prime and n is some positive integer. Later on we will also construct such fields for different p and n. So far we have only seen $F = \mathbb{F}_p (= \mathbb{Z}_p)$, that is for the case $n = 1$.

To prove that $|F| = p^n$ we need the concept of vector spaces.

Example 6.33. Let F be a field and K a subfield of F (that is, $K \subseteq F$ and K is itself a field). Then F is a vector space over K: check the axioms. Now $u, v, w \in F$ and $\alpha, \beta \in K \subseteq F$. Hence the axioms all follow from the fact that F is a field.

If F is finite, then F must have a finite basis $e_1, e_2, \dots, e_n \in F$ as a vector space over K. Then we have $F = \{\alpha_1 e_1 + \cdots + \alpha_n e_n \mid \alpha_i \in K\}$, which implies that $|F| = |K|^n$.

If we could always find a subfield of F of prime order, this would prove that $|F| = p^n$ for some prime p. It turns out that we can find such a K whenever F is finite. Let

$$\begin{aligned} K &= \{\, 1\,,\, 1+1\,,\, 1+1+1\,,\, 1+1+1+1\,,\, \dots \} \\ &= \{\, 1\,, 2\cdot 1\,,\, 3\cdot 1\,,\, 4\cdot 1\,,\, \dots \}. \end{aligned}$$

If F is finite, then there are numbers $r < s$ with $r \cdot 1 = s \cdot 1$, and so $(s - r) \cdot 1 = 0$. Hence there is some positive integer m with $m \cdot 1 = 0$. The smallest such m must be prime, because if $m = a \cdot b$ with $1 < a, b < m$, then

$$m \cdot 1 = (a \cdot 1)(b \cdot 1)$$

and as F has no zero divisors, either $a \cdot 1 = 0$ or $b \cdot 1 = 0$, contradicting the minimality of m.

We conclude the following:

Theorem 6.34. *Let F be a finite field. Then $|F| = p^n$ for some prime p and some positive integer n.*

This prime p is called the *characteristic* of the field F. It is also true that for each prime power p^n there exists a field F with $|F| = p^n$. In fact it is even a unique field (up to isomorphism). We will not prove this but we will look at an example that gives an idea of how to construct these fields.

Example 6.35. Let $\mathbb{Z}_5[x]$ be the set of all polynomials in x with coefficients in $\mathbb{Z}_5$. (This is actually a ring.) Now let us say that $f, g \in \mathbb{Z}_5[x]$ are equivalent (denoted $[f] = [g]$) if $(x^2 + 2) \mid (f(x) - g(x))$. Write also $h(x) = x^2 + 2$.

Now $F = \{[a + bx] \mid a, b \in \mathbb{Z}_5\}$ gives the full set of equivalence classes since any $f(x)$ has remainder of degree ≤ 1 after division by $h(x) = x^2 + 2$.

We turn F into a ring by defining the multiplication

$$\begin{aligned} [ax + b] \cdot [cx + d] &= [acx^2 + (ad + bc)x + bd] = [acx^2 - ac(x^2 + 2) + (ad + bc)x + bd] \\ &= [(ad + bc)x + bd - 2ac]. \end{aligned}$$

The ring F is a field since $[ax + b]$ has inverse $[cx + d]$ if and only if

$$\begin{cases} ad + bc = 0 \\ bd - 2ac = 1 \end{cases} \iff \begin{pmatrix} b & a \\ -2a & b \end{pmatrix} \begin{pmatrix} c \\ d \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix},$$

and this system has a unique solution (c, d), since the determinant equals $b^2 + 2a^2$ which is always non-zero, as at least one of a or b is non-zero.

We have $|F| = 5^2$.

In general, we denote such a set F by $\mathbb{Z}_n[x]/(h(x))$ and any $\mathbb{Z}_n[x]/(h(x))$ is a ring. Furthermore, we get a field if and only if n is prime and $h(x)$ cannot be factorised in $\mathbb{Z}_n[x]$. (More on this will be covered in the course Algebraic Structures.)

7. Coding

7.1. **Introduction.** In more general terms, coding means representing information in a different form, for example:

- turning speech into writing;
- translating from one language to another;
- the Morse code: translating single letters into conventional sequences of dots and lines (representing short and long impulses);
- the postcode: representing a location by a few letters and/or numbers;
- stenography (the process of writing in shorthand, for speed).

Of course some of these are rather generic examples of coding. More mathematically interesting types of coding are used in modern communication technology. Here are some of the most important types of codes:

- *data compression codes*: shrinking information for quicker transmission, sometimes with a loss of information;
- *cryptographic codes*: turning information into a form which only the intended receiver can understand;
- *error-correcting codes*: adding some redundant information which allows one to recover from possible transmission errors.

Often several of these are combined in a given application, but in this course we are largely interested in error-correcting codes. As the above three general types of codes aim at different goals, they also display the following differences: data compression reduces the size of information (possibly a lot); cryptography generally leaves the size almost unchanged; error-correcting codes expand the size (but as little as possible).

Sometimes it is enough that a code *detects* errors, even without being able to correct them. This is good enough in situations where it would be possible and not too costly to simply ask for the message to be retransmitted if an error is detected. Consider for example the International Standard Book Number ISBN described below. (More precisely, we describe here ISBN-10, which was used until 2006. Nowadays one uses ISNB-13.)

Any book has a unique code, e.g. the book 'Coding Theory' by San Ling and Chaoping Xing is listed as 0-521-52923-9. The last digit of this code is a check digit: if the previous digits are $x_1, x_2, \ldots, x_9$, then the last digit is compute as

$$x_{10} = 1 \cdot x_1 + 2 \cdot x_2 + 3 \cdot x_3 + \cdots + 9 \cdot x_9 \pmod{11},$$

where an 'X' is used when the remainder modulo 11 equals 10. In our example,

$$9 \equiv 1 \cdot 0 + 2 \cdot 5 + 3 \cdot 2 + 4 \cdot 1 + 5 \cdot 5 + 6 \cdot 2 + 6 \cdot 2 + 7 \cdot 9 + 8 \cdot 2 + 9 \cdot 3 \pmod{11}.$$

Suppose I made an error when copying the code, e.g. I confused the 1 in the fourth position with a 7. Then the sum above would give

$$0 \equiv 1 \cdot 0 + 2 \cdot 5 + 3 \cdot 2 + 4 \cdot \mathbf{7} + 5 \cdot 5 + 6 \cdot 2 + 6 \cdot 2 + 7 \cdot 9 + 8 \cdot 2 + 9 \cdot 3 \pmod{11},$$

so we know that there is an error. But the code is even better, suppose I swapped the 2 and 3 in the eighth and ninth position, then again my sum will give a different value as the coefficients are different:

$$8 \equiv 1 \cdot 0 + 2 \cdot 5 + 3 \cdot 2 + 4 \cdot 1 + 5 \cdot 5 + 6 \cdot 2 + 6 \cdot 2 + 7 \cdot 9 + 8 \cdot \mathbf{3} + 9 \cdot \mathbf{2} \pmod{11}.$$

Again, the ISBN code does not tell us where the error is; also if we made two errors it may well be that the sum equals the check digit.

Another such example is the Swedish personal identification numbers, which have 10 digits where the last is a control digit computed from the previous nine as follows. For 750327484 the digits are multiplied alternatingly by 2 and 1 starting with 2 any number ≥ 10 is replaced by the sum of its digits:

$$\begin{array}{cccccccccc} 7 & 5 & 0 & 3 & 2 & 7 & 4 & 8 & 4 & x \\ 2 & 1 & 2 & 1 & 2 & 1 & 2 & 1 & 2 & 1 \\ 5 & 5 & 0 & 3 & 4 & 7 & 8 & 8 & 8 & x \end{array}$$

Next, we sum them:

$$5 + 5 + 0 + 3 + 4 + 7 + 8 + 8 + 8 + x = 48 + x$$

Then x is chosen so that the sum $\equiv 0 \pmod{10}$. In this case $x = 2$, so the full number is

$$\begin{array}{cccccccccc} 7 & 5 & 0 & 3 & 2 & 7 & 4 & 8 & 4 & 2 \end{array}$$

If one digit is wrong we always get a sum $\not\equiv 0 \pmod{10}$, so this error is detected.

Moreover if two consecutive digits are swapped, we get the contribution $a + 2b$, instead of $b + 2a$, to the sum and they are different modulo 10, unless $a \equiv b$, that is $a = b$. (There is the exception that swapping consecutive 09 is not detected; this has to do with replacing a number with its digit sum if it is greater than 9.)

If two (e.g. non-consecutive) digits are wrong, we can get a new valid personal identification number, for example:

$$7503274842 \quad \text{and} \quad 7\mathbf{4}0\mathbf{4}274842$$

are both valid.

From now onwards, we will assume that our symbols come from a finite field F and that the code words are symbol strings of some fixed length n.

Definition 7.1. An injective map $f : F^m \to F^n$, for $m < n$, is called a *coding function* and its image $f(F^m) = C$ is called a *code*.

Example 7.2. Let $F = \mathbb{Z}_7$ and define $f : \mathbb{Z}_7^3 \to \mathbb{Z}_7^4$ by

$$(x_1, x_2, x_3) \longmapsto (x_1, x_2, x_3, x_1 + x_2 + x_3).$$

As f is injective (exercise), we have that f is a coding function. The code $C = f(\mathbb{Z}_7^3)$ contains $7^3 = 343$ words. One of them is, for example, the word $f((0, 1, 6)) = (0, 1, 6, 0)$, but for instance $(0, 1, 6, 1)$ is not a word in the code.

Example 7.3. Let $F = \mathbb{Z}_2[x]/(x^2 + x + 1)$ be a field of four elements. Define $g : F^2 \to F^3$ by

$$(f_1, f_2) \longmapsto (f_2, x f_1, f_1 - f_2).$$

More specifically, writing $f_1 = ax + b$ and $f_2 = cx + d$, for $a, b, c, d \in \mathbb{Z}_2$, we have

$$\begin{aligned} g((ax+b, cx+d)) &= (cx+d, ax^2+bx, (a-c)x+(b-d)) \\ &= (cx+d, (ax+a)+bx, (a+c)x+(b+d)) \\ &= (cx+d, (a+b)x+a, (a+c)x+(b+d)). \end{aligned}$$

This map g is injective (exercise), and if we take, for example, $a = c = 1$ and $b = d = 0$, then $g((x, x)) = (x, x^2, 0) = (x, x + 1, 0)$ is a code word in $C = g(F^2)$.

Next we will introduce a distance function between code words. This distance function is chosen such that words of the code are far apart, so that a few incorrect letters do not turn one code word into another.

Definition 7.4. The *Hamming distance* $d(\mathbf{x}, \mathbf{y})$ between $\mathbf{x} = (x_1, \ldots, x_n)$ and $\mathbf{y} = (y_1, \ldots, y_n)$ in F^n is the number of coordinates i with $x_i \neq y_i$.

Example 7.5. For $F = \mathbb{Z}_{13}$ and $n = 4$, we have $d(0123, 1121) = 2$.

For $F = \mathbb{Z}_2[x]/(x^2 + x + 1)$ with $n = 3$, we have $d\big([x][x + 1][1], [x][x + 7][0]\big) = 1$.

Definition 7.6. Let C be a code in F^n. Then $d(C) = \min\{d(\mathbf{x}, \mathbf{y}) \mid \mathbf{x}, \mathbf{y} \in C, \mathbf{x} \neq \mathbf{y}\}$ is called the *separation* of the code.

Example 7.7. Consider $\mathbb{Z}_2^3$, and create a code containing exactly 2 words with maximal separation. And similarly for exactly 3 words and exactly 4 words. (One can illustrate the distance between elements in $\mathbb{Z}_2^3$ with a cube. The distance between two words is the number of edges on the shortest path between them.)

Solution: The maximum distance between words is 3. For example $C_1 = \{000, 111\}$ is a code of separation 3 and this is the best separation we can obtain for two words.

For three words: if we start with two words of distance 3 every other word will be just one step from one of the selected words. This shows that $d(C)$ cannot be 3.

What about $d(C) = 2$? This can be done, as seen below. Let us begin with two words of distance 2, for instance 000 and 101. Then we still have two words of distance two to both of the selected words: 110 and 011. Thus $C_2 = \{000, 101, 110\}$ has separation 2. Adding the other of these two words we obtain a four word code without lowering the separation: $C_3 = \{000, 101, 110, 011\}$ with $d(C_3) = 2$.

Consider $C_3 = \{000, 101, 110, 011\} \subseteq \mathbb{Z}_2^3$ from above. If we receive a word with one incorrect symbol, for example 001, we know something is wrong because the word is not a code word. We need to make at least two mistakes (2 equals the separation of C) to transform a code word into a new code word. We know 001 is incorrect but we cannot say what it should have been: 000, 101 and 011 are all equally likely candidates.

If we consider $C_1 = \{000, 111\}$ and we get one symbol wrong then we can correct the word. For instance, 101 must come from 111.